

CODING PRACTICE & PROGRESS TRACKER

Project Documentation

B.Tech Computer Science Engineering Mini Project

Project Type: Web Application

Purpose: Coding practice, progress monitoring and skill improvement

1. Abstract

Coding Practice & Progress Tracker is a web application designed to help students improve their programming and problem-solving skills. The system provides coding problems, daily challenges, progress tracking, goals, achievements, and a leaderboard. Students can monitor their coding activity and maintain consistent practice through a simple and interactive dashboard.

2. Introduction

Students often practice coding on different platforms without having a single place to monitor their learning progress. This project provides an organized platform where students can practice coding, record solved problems, track streaks, set goals, and view their overall improvement.

3. Problem Statement

Students may find it difficult to maintain consistent coding practice and measure their progress. Existing practice activities are often scattered across multiple platforms. The proposed system provides a centralized web application for coding practice and progress tracking.

4. Objectives

- Provide coding problems categorized by programming language and difficulty.
- Encourage daily coding practice through challenges and streaks.
- Track solved problems, practice time and learning progress.
- Allow students to set and monitor coding goals.
- Provide achievements and a leaderboard to increase motivation.

5. Proposed System

The proposed system provides a personalized coding practice environment. Users can create an account, access coding problems, mark problems as solved, complete daily challenges, set goals, earn badges, and monitor progress through charts and statistics.

6. Main Modules

User Authentication: Registration, login, logout and profile management.

Dashboard: Displays solved problems, streak, practice hours and progress.

Coding Practice: Problems categorized by language, topic and difficulty.

Daily Challenge: Provides a coding challenge every day.

Progress Tracker: Shows daily, weekly and monthly coding statistics.

Goals: Allows users to set and monitor coding targets.

Achievements: Awards badges for completing milestones.

Leaderboard: Ranks users based on coding activity and points.

Notes: Allows students to save coding-related notes.

7. Technologies

Frontend: HTML, CSS, JavaScript / modern web UI framework

Backend: Base44 application services and database

Database: Application database for users, problems and progress

Development Tool: Base44 AI

Platform: Responsive web application

8. Functional Requirements

The system should allow users to register and log in, browse coding problems, search and filter problems, mark problems as solved, complete daily challenges, track coding streaks, create goals, earn achievements, view progress charts, manage notes, and view leaderboard rankings.

9. Non-Functional Requirements

The application should be responsive, user-friendly, reliable, secure, easy to maintain, and provide fast navigation. The interface should work effectively on desktop, tablet, and mobile devices.

10. Database Entities

User: Stores student profile information.

Problem: Stores coding questions, topics, languages and difficulty.

Submission/Progress: Stores solved and attempted problems.

Goal: Stores user coding targets and completion status.

Achievement: Stores badges earned by users.

Challenge: Stores daily coding challenges.

Note: Stores personal coding notes.

11. Advantages

- Encourages regular coding practice.
- Provides a centralized progress dashboard.
- Helps students identify strengths and areas for improvement.
- Motivates users through goals, streaks, badges and rankings.
- Suitable for B.Tech CSE students and beginners.

12. Future Enhancements

Future versions can include an online code compiler, automated test-case evaluation, AI-based coding hints, personalized problem recommendations, integration with coding platforms, email notifications, advanced analytics, and a mobile application.

13. Conclusion

Coding Practice & Progress Tracker provides a simple and effective solution for students who want to develop consistent coding habits. By combining coding practice with progress tracking, goals, challenges and achievements, the application helps students improve their programming skills and monitor their learning journey.

End of Project Documentation